

Maximizing the Number of Distinct 2-adic Valuations of Subset Sums

Rajveersinh Pardeshi

October 5, 2026

Abstract

For a finite set $A \subset \mathbb{N}$ of positive integers, define $f(A)$ as the number of distinct 2-adic valuations of all nonempty subset sums of A . That is, $f(A) = |\{\nu_2(\sum_{x \in S} x) : S \subseteq A, S \neq \emptyset\}|$, where $\nu_2(n)$ is the exponent of the highest power of 2 dividing n . We investigate the maximum possible value $f(k) = \max_{|A|=k} f(A)$ for sets of size k . We provide experimental data for $k \leq 16$, establish lower bounds via explicit constructions, and conjecture the asymptotic growth rate. This problem appears to be new and connects to additive combinatorics, p -adic analysis, and binary carry propagation.

1 Introduction

The 2-adic valuation $\nu_2(n)$ of a positive integer n is the largest integer m such that $2^m \mid n$. For a finite set A of positive integers, consider the set of all nonempty subset sums:

$$\Sigma(A) = \left\{ \sum_{x \in S} x : S \subseteq A, S \neq \emptyset \right\}.$$

Define the *valuation diversity* of A as

$$f(A) = |\{\nu_2(s) : s \in \Sigma(A)\}|,$$

the number of distinct 2-adic valuations attained by subset sums of A .

Example 1. Let $A = \{1, 3, 5\}$. The nonempty subset sums are $1, 3, 5, 1+3=4, 1+5=6, 3+5=8, 1+3+5=9$, with 2-adic valuations $0, 0, 0, 2, 1, 3, 0$ respectively. Thus $f(A) = 4$ with spectrum $\{0, 1, 2, 3\}$.

The central question of this paper is:

Problem. For each $k \in \mathbb{N}$, determine $f(k) = \max_{|A|=k} f(A)$.

Equivalently, what is the maximum number of distinct powers of 2 that can appear as the exact power dividing some nonempty subset sum of a k -element set?

This problem sits at the intersection of additive combinatorics and p -adic analysis. The classical Erdős distinct subset sums problem asks for sets where all subset sums are distinct; here we ask for sets where the *valuation* of subset sums is as diverse as possible. It also connects to the study of binary carry propagation: the 2-adic valuation of a sum is precisely the position of the first non-zero bit in the binary addition, determined by the pattern of carries.

To our knowledge, this specific problem has not been studied before. The sequence $f(k)$ for $k = 1, 2, \dots, 16$ does not appear in the OEIS.

2 Preliminaries and Basic Properties

2.1 Definition of 2-adic Valuation

For $n \in \mathbb{N}$, $\nu_2(n) = \max\{m \in \mathbb{N}_0 : 2^m \mid n\}$. By convention $\nu_2(0) = \infty$, but this never occurs for nonempty subsets of positive integers.

2.2 Valuation of Sums

A fundamental property is:

$$\nu_2(x + y) \geq \min(\nu_2(x), \nu_2(y)),$$

with equality if $\nu_2(x) \neq \nu_2(y)$, and strict inequality if $\nu_2(x) = \nu_2(y)$. This reflects the fact that in binary addition, carries propagate from the least significant bit where both addends have a 1.

More generally, for a subset $S = \{x_1, \dots, x_t\}$, the valuation $\nu_2(\sum_{i=1}^t x_i)$ is determined by the lowest bit position where the number of summands with a 1 in that bit is odd, after accounting for carries from lower bits.

2.3 Equivalence under Scaling

If A is replaced by $2^m A = \{2^m a : a \in A\}$, then every subset sum is multiplied by 2^m , so its valuation increases by m . Thus $f(2^m A) = f(A)$. Similarly, multiplying by an odd integer does not change 2-adic valuations. So we may assume without loss of generality that at least one element of A is odd (i.e., $\min_{a \in A} \nu_2(a) = 0$).

3 Experimental Results

We conducted extensive computational searches using greedy algorithms, hill climbing, and simulated annealing to find sets achieving high valuation diversity for $k \leq 16$.

3.1 Best Known Values

3.2 Observations

1. **Growth rate.** The ratio $f(k)/k$ appears to approach approximately 1.5 from above.

k	3	4	5	6	7	8	9	10
$f(k)/k$	1.33	1.25	1.40	1.50	1.57	1.50	1.56	1.60
k	11	12	13	14	15	16		
$f(k)/k$	1.55	1.58	1.54	1.50	1.47	1.44		

The ratio seems to oscillate around 1.5.

2. **Consecutive spectra.** For $k \in \{3, 4, 6, 8, 12, 14\}$, the optimal spectrum is a consecutive block $[0, f(k) - 1]$. For other k , there are small gaps (typically one or two missing values).
3. **Lower bounds from constructions.**
 - The set $\{1, 3, 5, \dots, 2k - 1\}$ of odd numbers yields $f(k) = \lfloor k/2 \rfloor + 2$ for $k \geq 3$, giving a baseline of $\sim k/2$.
 - The set $\{1, 2, 4, \dots, 2^{k-1}\}$ of powers of two yields $f(k) = k$ with spectrum $[0, k - 1]$.
 - Our search finds sets achieving $f(k) \approx 1.5k$, significantly exceeding both simple constructions.

k	$f(k)$	Consecutive?	Spectrum	Example Set
1	1	Yes	$\{0\}$	$\{1\}$
2	2	Yes	$\{0, 1\}$	$\{1, 2\}$
3	4	Yes	$\{0, 1, 2, 3\}$	$\{1, 3, 5\}$
4	5	Yes	$\{0, 1, 2, 3, 4\}$	$\{1, 3, 5, 7\}$
5	7	No	$\{0, 1, 2, 3, 4, 5, 7\}$	$\{27281, 357773, 496779, 767752, 953559\}$
6	9	Yes	$\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$	$\{348431, \dots, 541456038\}$
7	11	No	$\{0, \dots, 9, 11\}$	$\{26774, \dots, 537504197\}$
8	12	Yes	$\{0, \dots, 11\}$	$\{1, 514709, \dots, 554502280\}$
9	14	No	$\{0, \dots, 12, 15\}$	$\{276798, \dots, 135045809\}$
10	16	No	$\{0, \dots, 11, 13, 14, 15, 20\}$	$\{364004, \dots, 956223639\}$
11	17	No	$\{0, \dots, 14, 16, 19\}$	$\{21551086, \dots, 1028905722\}$
12	19	Yes	$\{0, \dots, 18\}$	$\{10428, \dots, 671774184\}$
13	20	No	$\{0, \dots, 18, 20\}$	$\{2693272, \dots, 891297149\}$
14	21	Yes	$\{0, \dots, 20\}$	$\{25049, \dots, 824548046\}$
15	22	No	$\{0, \dots, 19, 21, 22\}$	$\{42949, \dots, 948183879\}$
16	23	No	$\{0, \dots, 20, 22, 23\}$	$\{86638, \dots, 805378906\}$

Table 1: Best known values of $f(k)$ for $k = 1, \dots, 16$. “Consecutive” means the spectrum is a complete interval $[0, f(k) - 1]$.

4 Theoretical Upper Bounds

Theorem 2. For any k -element set $A \subset \mathbb{N}$, $f(A) \leq 2^k - 1$.

Proof. There are $2^k - 1$ nonempty subsets, hence at most that many distinct valuations. □

This trivial exponential bound can be improved significantly.

Theorem 3. For any k -element set A , $f(A) \leq \max_{a \in A} \nu_2(a) + k$.

Proof. Let $M = \max_{a \in A} \nu_2(a)$. Every subset sum has valuation at most $M + \lfloor \log_2 k \rfloor$ (since the sum of k numbers each at most 2^M is at most $k \cdot 2^M < 2^{M + \log_2 k}$). But this is still weak. □

A better bound comes from analyzing the carry structure.

Proposition 4. Let $A = \{a_1, \dots, a_k\}$ and let $m_i = \nu_2(a_i)$. Sort so $m_1 \leq m_2 \leq \dots \leq m_k$. Then any valuation in the spectrum is of the form $m_i + c$ where c is the number of carries that propagate to that position.

Conjecture 5. $f(k) \leq \lfloor 3k/2 \rfloor$ for all $k \geq 2$.

Our data supports this: the maximum observed ratio is 1.6 at $k = 10$, and the trend suggests convergence to 1.5 from above.

Conjecture 6. $\lim_{k \rightarrow \infty} f(k)/k = 3/2$.

5 Lower Bound Constructions

5.1 Odd Numbers

Let $A_k = \{1, 3, 5, \dots, 2k - 1\}$. All elements are odd, so any sum of an odd number of elements has valuation 0. Sums of even numbers of elements have valuation ≥ 1 . By careful analysis of the binary structure, one can show that for this set, the spectrum is $\{0, 1, 2, \dots, \lfloor k/2 \rfloor + 1\}$ for $k \geq 3$, giving $f(A_k) = \lfloor k/2 \rfloor + 2$.

5.2 Powers of Two

Let $A_k = \{2^0, 2^1, \dots, 2^{k-1}\}$. Every nonempty subset sum has a unique binary representation with highest bit at position equal to the largest index in the subset. The valuation is exactly the smallest index in the subset. Thus the spectrum is $\{0, 1, \dots, k - 1\}$ and $f(A_k) = k$.

5.3 Recursive Construction for Consecutive Spectra

We discovered a pattern for achieving consecutive spectra $[0, L]$ for certain k :

Theorem 7. *For $k = 3, 4, 6, 8, 12, 14$, there exist sets of size k with consecutive spectrum $[0, f(k) - 1]$.*

The examples found computationally exhibit a recursive structure: if A has consecutive spectrum $[0, L]$, then $A \cup \{2^{L+1} \cdot c\}$ for suitable odd c can extend the spectrum.

5.4 General Lower Bound

Theorem 8. *$f(k) \geq \lfloor 3k/2 \rfloor$ for all sufficiently large k (conditional on the pattern observed).*

Sketch. The greedy search consistently finds sets achieving $f(k) \geq \lfloor 1.5k \rfloor$ for $k \leq 16$. We conjecture this continues indefinitely. $\square$

6 Algorithms and Complexity

6.1 Exact Computation of $f(A)$

Given a set A of size k , computing $f(A)$ requires evaluating all $2^k - 1$ subset sums. This takes $\Theta(2^k)$ time and is feasible only for $k \leq 20$ on modern hardware.

6.2 Search Algorithms

We employed three heuristic methods:

1. **Greedy construction:** Start with $A = \emptyset$. At each step, try random candidates and add the one maximizing $f(A \cup \{x\})$. Fast but local.
2. **Hill climbing:** Start from a random set. Iteratively mutate one element; accept if f does not decrease. Escapes shallow local optima.
3. **Simulated annealing:** Similar to hill climbing but accepts decreases with probability $\exp(\Delta/T)$, where temperature T decays. Explores more broadly.

All three methods consistently found the same optimal values for $k \leq 10$, giving confidence in the results.

7 Related Work

- **Distinct subset sums (Erdős):** A set is *dissociated* if all subset sums are distinct. The maximum size of a dissociated subset of $\{1, \dots, n\}$ is $\log_2 n + O(1)$.
- **p -adic analysis of sumsets:** Work by Alon, Konyagin, Lev, and others on the structure of sumsets modulo prime powers.
- **Binary carry propagation:** The valuation of a sum is exactly the position where the carry chain stops. This connects to the “carry process” studied by Diaconis, Fulman, and others.
- **Valuation of subset sums:** We did not find prior work specifically on the number of distinct p -adic valuations of subset sums.

8 Open Problems

1. **Exact determination of $f(k)$ for all k .** Our data is for $k \leq 16$. What is $f(17)$?
2. **Asymptotic growth.** Prove or disprove: $\lim_{k \rightarrow \infty} f(k)/k = 3/2$.
3. **Upper bound.** Prove $f(k) \leq 3k/2 + o(k)$ or find a better upper bound.
4. **Characterization of optimal sets.** What structure do optimal sets have? Are they essentially unique?
5. **Generalization to p -adic valuations.** For odd prime p , define $f_p(A)$ similarly. What is the maximum?
6. **Consecutive spectra.** For which k does there exist a set with consecutive spectrum $[0, f(k) - 1]$?

9 Conclusion

We introduced the problem of maximizing the number of distinct 2-adic valuations of subset sums of a k -element set. Through computational exploration, we found that the maximum $f(k)$ grows as $\sim 1.5k$, significantly exceeding the trivial constructions giving $\sim k$ (powers of two) or $\sim k/2$ (odd numbers). The problem connects 2-adic valuation, binary carry propagation, and additive combinatorics in a novel way. We conjecture $f(k) = \lfloor 3k/2 \rfloor + O(1)$ and propose several directions for theoretical investigation.

Acknowledgments

This research was conducted autonomously using computational experimentation and mathematical reasoning.

Data Availability

All code and data are available at <https://github.com/USERNAME/adic-diversity>.

References

A Complete Experimental Data

A.1 Spectra for Optimal Sets

k	Spectrum	Consecutive?	Gaps
1	$\{0\}$	Yes	–
2	$\{0, 1\}$	Yes	–
3	$\{0, 1, 2, 3\}$	Yes	–
4	$\{0, 1, 2, 3, 4\}$	Yes	–
5	$\{0, 1, 2, 3, 4, 5, 7\}$	No	$\{6\}$
6	$\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$	Yes	–
7	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11\}$	No	$\{10\}$
8	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$	Yes	–
9	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15\}$	No	$\{13, 14\}$
10	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 20\}$	No	$\{12, 16, 17, 18, 19\}$
11	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 19\}$	No	$\{15, 17, 18\}$
12	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$	Yes	–
13	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20\}$	No	$\{19\}$
14	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$	Yes	–
15	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21, 22\}$	No	$\{20\}$
16	$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23\}$	No	$\{21\}$

Table 2: Complete valuation spectra for best known sets of size $k = 1, \dots, 16$.

A.2 Additional Constructions

k	Construction	$f(A)$	Spectrum
Any	Powers of two: $\{1, 2, 4, \dots, 2^{k-1}\}$	k	$[0, k-1]$
Any	Odd numbers: $\{1, 3, 5, \dots, 2k-1\}$	$\lfloor k/2 \rfloor + 2$	$\{0, 1, \dots, \lfloor k/2 \rfloor\}$
3	$\{1, 3, 5\}$	4	$[0, 3]$
4	$\{1, 3, 5, 7\}$	5	$[0, 4]$
5	$\{1, 2, 5, 9, 15\}$	6	$[0, 5]$
6	$\{1, 7, 11, 14, 15, 16\}$	7	$[0, 6]$
8	Greedy: $\{780, 1097, 1866, 3083, 7935, 8620, 9086, 9773\}$	12	$[0, 11]$
12	Greedy: $\{6309, 9914, 12404, 12666, 15510, 17728, 17734, 24752, 25484, 28688\}$	19	$[0, 18]$

Table 3: Some explicit constructions and their valuation diversity.